DO NOW · AO1 REVISION

Random and systematic errors

Working scientifically
Both papers · Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe the effect of a random error on a set of readings.**	It scatters the readings unpredictably.	☐
2 ★	**Describe the effect of a systematic error on a set of readings.**	It shifts every reading in the same direction by the same amount.	☐
3	**Give an example of a random error.**	Reaction time when using a stopwatch.	☐
4 ★	**Give an example of a systematic error.**	A balance that does not read zero when empty.	☐
5 ★	**State how the effect of random error can be reduced.**	Take more repeats and calculate a mean.	☐
6	**Explain why repeating readings does not remove a systematic error.**	Every reading is shifted in the same direction, so the mean is shifted too.	☐
7	**State how a systematic error can be found and corrected.**	Check the zero and calibration of the instrument, then correct the method.	☐
8	**Explain why "human error" is not a good answer when evaluating an experiment.**	It is too vague: you must name what actually went wrong.	☐
9	**A ruler is read from the wrong zero mark every time. State the type of error this causes.**	A systematic error	☐
10	**Does better resolution remove a systematic error?**	No: the readings are finer but still shifted.	☐
11	**A graph should be a straight line through the origin, but the line of best fit misses the origin. Name the type of error this suggests.**	A systematic error (such as a zero error): every reading is shifted by the same amount.	☐

© 2026 PanPhy Labs. All rights reserved.